Unit

21 Oxidation and reduction in chemical cells

Pre-study
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Redox reactions in various chemical cells
- Hydrogen–oxygen fuel cells

21.1 Oxidation and reduction in a simple chemical cell

Fig. 21.1 shows a chemical cell. The reaction in this simple chemical cell is an example of a redox reaction.

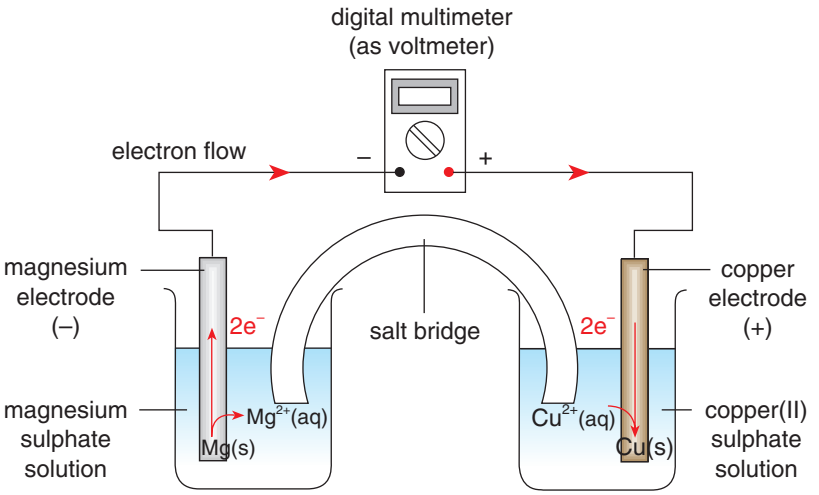


Fig. 21.1 Redox reaction in a simple chemical cell

Oxidation occurs at the magnesium electrode (negative electrode or **anode**):

$$Mg(s) \longrightarrow Mg^{2+}(aq) + 2e^-$$

Reduction occurs at the copper electrode (positive electrode or **cathode**):

$$Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$$

The overall equation is:

$$Mg(s) + Cu^{2+}(aq) \longrightarrow Mg^{2+}(aq) + Cu(s)$$

Do you know

Chemical cell mnemonic

Remember our two friends LEO and GER?

LEO is An Ox (**An**ode — **Ox**idation);



GER is a Red Cat (**Cat**hode — **Red**uction).

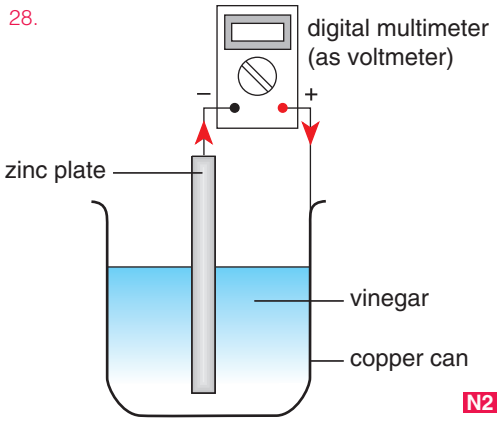


anode 陽極　　cathode 陰極

Example 21.1

Q Consider the chemical cell shown below.

N1 Refer to Teacher's Resources Book page 28.



N2

Electrons flowed from the zinc plate to the copper can in the external circuit. Colourless gas bubbles were given off on the inner wall of the copper can.

a) i) Write an ionic half-equation for the reaction that occurred at the zinc plate.

ii) What would happen to the mass of the zinc plate?

b) Write an ionic half-equation for the reaction that occurred on the inner wall of the copper can.

c) It was found that colourless gas bubbles were also given off at the surface of the zinc plate immersed in the vinegar. Explain this observation.

CE 2007 Paper 1 Q4(b),(c)
CE 2009 Paper 2 Q9

A a) i) $Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$

ii) The mass of the zinc plate would decrease.

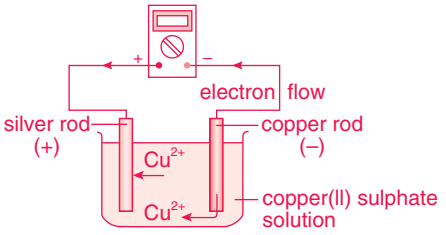
b) $2H^+(aq) + 2e^- \longrightarrow H_2(g)$

c) Zinc reacted directly with vinegar to give hydrogen gas.

N2 Change of Colour intensity of the electrolyte of a chemical cell

Examination questions may ask about the change of the colour intensity of the electrolyte of a chemical cell, i.e. the change of concentration of coloured ions in the electrolyte.

Copper rod is the anode where oxidation occurs.
$Cu(s) \longrightarrow Cu^{2+}(aq) + 2e^-$
Silver rod is the cathode where reduction occurs.
$Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$



The net effect is the transfer of copper from the copper rod to the silver rod. The concentration of copper(II) ions in the electrolyte remains the same. The blue colour of the solution does not change.

Practice 21.1

P21.1 Refer to Teacher's Resources Book page 49.

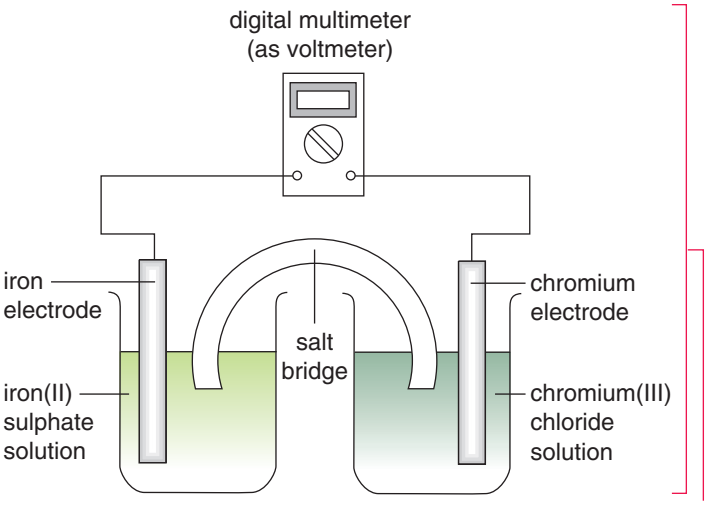
The diagram on the right shows a simple chemical cell. As time goes by, the colour of the iron(II) sulphate solution gradually fades out.

a) Write ionic half-equations for the reactions that occur at the two electrodes respectively.

b) State the direction of electron flow in the external circuit.

CE 2011 Paper 1 Q4(a),(b)

c) Explain whether the chromium electrode is the anode or the cathode of the chemical cell.



CE 2011 Paper 1 Q4

21.2 Redox reactions in a zinc-carbon cell

A zinc-carbon cell contains a carbon rod surrounded by a mixture of powdered carbon and manganese(IV) oxide (Figs. 21.2–21.3). The carbon rod is the positive electrode (or cathode). The outer zinc case is the negative electrode (or anode). The electrolyte is a moist paste of ammonium chloride.

CE 2006 Paper 2 Q32
CE 2007 Paper 2 Q44
CE 2011 Paper 2 Q32

N3 The zinc case becomes thinner when a zinc-carbon cell is being used.



Fig. 21.2 A zinc-carbon cell is cut into half to reveal its structure

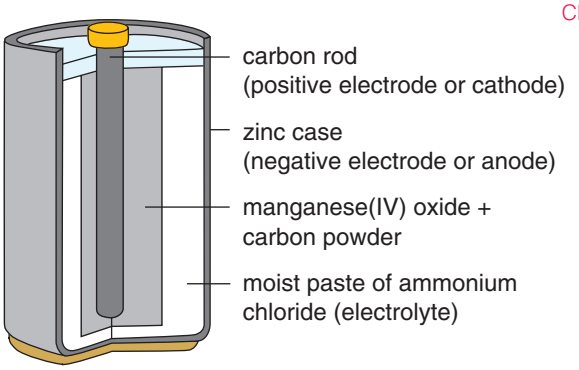


Fig. 21.3 The structure of a zinc-carbon cell

CE 2005 Paper 1 Q9(c)
CE 2009 Paper 2 Q34

N4 In a circuit using a zinc-carbon cell to supply electricity, electrons in the external circuit flow to the carbon electrode of the zinc-carbon cell.

Oxidation occurs at the negative electrode (or anode):

$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$ N3

Reduction occurs at the positive electrode (or cathode): N4

$2NH_4^+(aq) + 2e^- \longrightarrow 2NH_3(aq) + H_2(g)$

Hydrogen is produced and accumulated at the surface of the positive electrode. Since hydrogen is a poor conductor of electricity, the accumulation of hydrogen at the positive electrode may hinder further reactions and decrease the current of the cell.

Manganese(IV) oxide, an oxidizing agent, is used to remove the hydrogen.

$$2MnO_2(s) + H_2(g) \longrightarrow Mn_2O_3(s) + H_2O(l)$$

The ammonia is taken up by zinc ions.

$$Zn^{2+}(aq) + 2NH_3(aq) + 2Cl^-(aq) \longrightarrow Zn(NH_3)_2Cl_2(s)$$

The overall equation is:

$$2MnO_2(s) + 2NH_4Cl(aq) + Zn(s) \longrightarrow Zn(NH_3)_2Cl_2(s) + Mn_2O_3(s) + H_2O(l)$$

There are two main disadvantages associated with this cell:

1 If a current is drawn from the cell rapidly, the gaseous product cannot be removed fast enough. The voltage drops as a result.

DSE 2014 Paper 1A Q16

2 The cell has a relatively short shelf life. There is a slow direct reaction between the zinc electrode and ammonium ions. After some time, the zinc case becomes too thin and the paste leaks out.

Reference Website

Refer to the following website for an animation illustrating the chemical reactions occurring in a zinc-carbon cell:
http://group.chem.iastate.edu/Greenbowe/sections/projectfolder/animations/ZnCbatteryV8web.html
(accessed July 2014)

21.3 Redox reactions in simple chemical cells with inert electrodes

Video

Redox reactions in simple chemical cells (using carbon electrodes)

In the half-cells described so far, the metal used as an electrode is also a reactant or a product in the redox reaction. However, not all half reactions involve a metal.

An **inert electrode** is required when the cell reaction involves only species in solution, not the electrode itself. An inert electrode does not chemically react in the cell reactions. Carbon and platinum electrodes are both inert electrodes.

inert electrode 惰性電極

Fig. 21.4 shows the set-up of a simple chemical cell. It is made up of two half-cells. Two carbon electrodes are put into separate beakers containing iron(II) sulphate solution and acidified potassium dichromate solution.

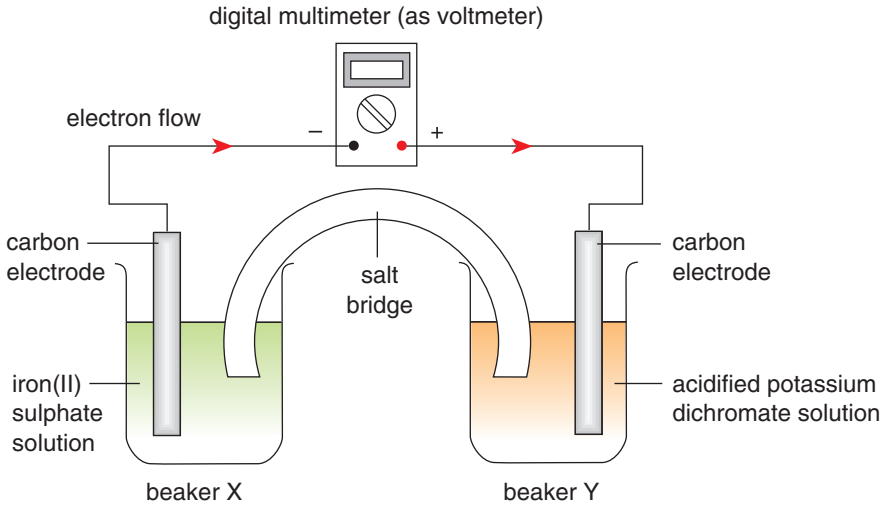


Fig. 21.4 A simple chemical cell with carbon electrodes N5

N5 Refer to Teacher's Resources Book page 29.

Look at the relative positions of the two relevant ionic half-equations in the electrochemical series (Fig. 21.5). As the dichromate ion is a stronger oxidizing agent than the iron(III) ion, it can oxidize the iron(II) ions to iron(III) ions.

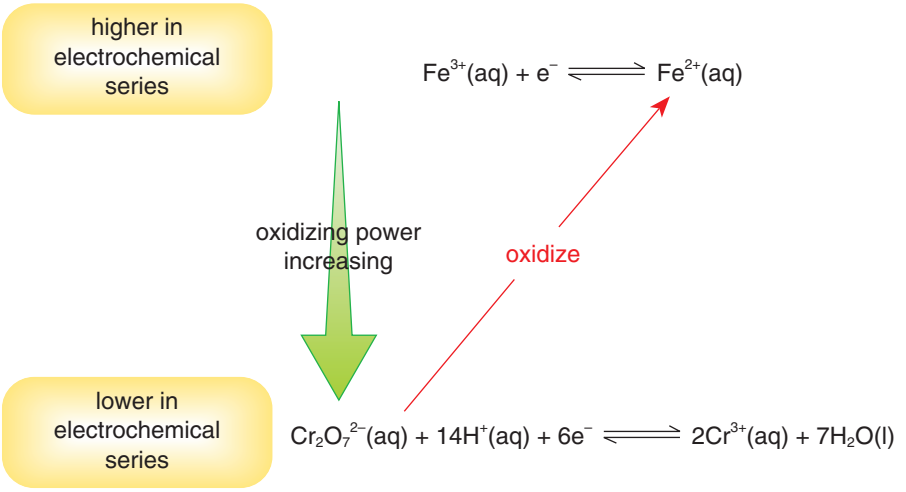


Fig. 21.5 The relative positions of the two ionic half-equations in the electrochemical series for the cell in Fig. 21.4

In beaker X, the solution changes from pale green to yellow-brown gradually. This is because the iron(II) ions are oxidized to iron(III) ions.

$$Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq) + e^-$$

In beaker Y, the solution changes from orange to green gradually. This is because the dichromate ions are reduced to chromium(III) ions.

$$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \longrightarrow 2Cr^{3+}(aq) + 7H_2O(l)$$

Electrons flow from the iron(II) sulphate solution to the acidified potassium dichromate solution in the external circuit.

The inert electrodes provide a surface for the reactions to occur.

Practice 21.2

P21.2 Refer to Teacher's Resources Book page 50.

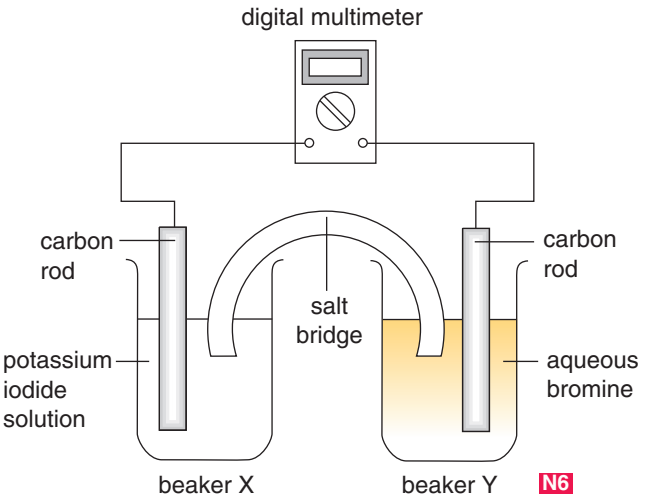
1 Consider the chemical cell shown on the right.

a) Compare the oxidizing power of bromine and iodine. Which is a stronger oxidizing agent?

b) Write ionic half-equations for the reactions that occur in beakers X and Y respectively.

c) State the observable changes for the mixtures in beakers X and Y respectively.

d) Suggest the direction of electron flow in the external circuit.



N6 Refer to Teacher's Resources Book page 30.

2 One type of breathalyzer for investigating drink-driving consists of a chemical cell. The breath of the driver is allowed to get into contact with one of the electrodes of the cell. Any ethanol in the breath would be converted to ethanoic acid and an electric current would be produced.

$$\underset{\text{ethanol}}{CH_3CH_2OH} \longrightarrow \underset{\text{ethanoic acid}}{CH_3COOH}$$

CE 2011 Paper 1 Q10(b)(i),(iii)

a) Explain whether ethanol undergoes oxidation or reduction when it is converted to ethanoic acid.

b) Decide whether the above-mentioned electrode is the anode or the cathode of the chemical cell.

c) Explain how this type of breathalyzer can estimate the amount of ethanol in the breath of the driver.

21.4 Fuel cells

The chemical cells we use everyday are small and portable. However, such types of cells can only contain a small amount of reactants which may be consumed in a short period of time. When one of the reactants is completely consumed, the cell can no longer generate a current. **Fuel cells** have no such limitation; the reactants (fuel and *oxidant*) can be supplied continuously to the cell from an external source (Fig. 21.6a–b).

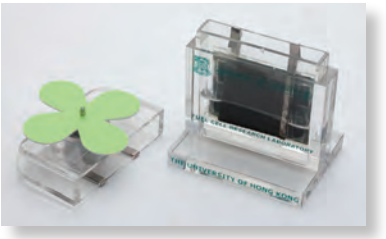


Fig. 21.6a A fan powered by a fuel cell using ethanol as fuel

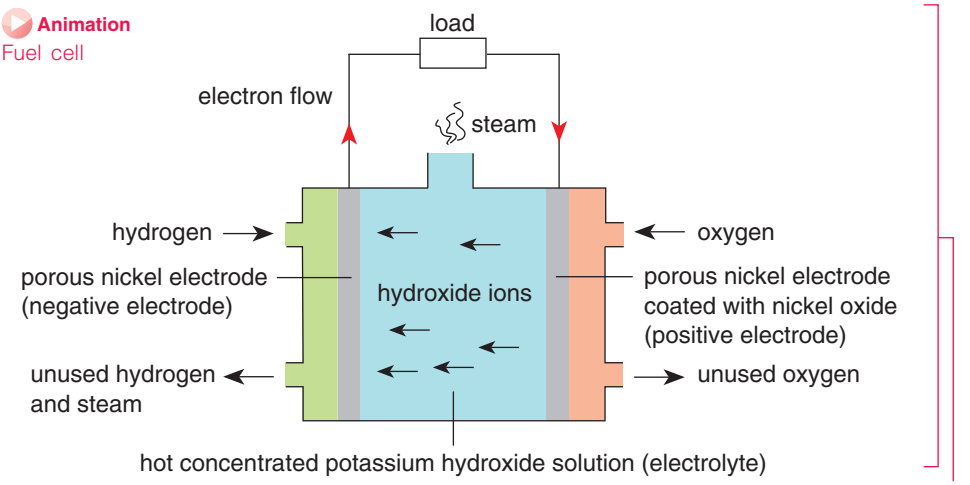


Fig. 21.6b A fuel cell-powered car

How does a hydrogen-oxygen fuel cell work?

There are many different types of fuel cells using different fuels, such as methanol. The **hydrogen-oxygen fuel cell** uses hydrogen as its fuel. Fig. 21.7 shows a simplified diagram of a hydrogen-oxygen fuel cell with an alkaline electrolyte. N6

Animation
Fuel cell



CE 2007 Paper 2 Q36

Fig. 21.7 Simplified diagram of a hydrogen-oxygen fuel cell

Do you know

Hydrogen-oxygen fuel cells for spacecraft

Hydrogen-oxygen fuel cells are used to provide the electrical energy for spacecraft. The fact that fuel cells are lightweight, compact and have no major moving parts makes them particularly suitable for this purpose. The water produced is used by the crew for drinking.

fuel cell 燃料電池　　oxidant 氧化劑　　hydrogen-oxygen fuel cell 氫氧燃料電池

In the cell, three compartments are separated from each other by two electrodes[1]. The fuel (hydrogen) is supplied into the negative electrode compartment while the oxidant (oxygen) is supplied into the positive electrode compartment. The electrolyte in the central compartment is hot concentrated potassium hydroxide solution.

DSE 2013 Paper 1B Q10(b)

At the negative electrode (or anode) compartment:

$$H_2(g) + 2OH^-(aq) \longrightarrow 2H_2O(l) + 2e^-$$

At the positive electrode (or cathode) compartment:

$$O_2(g) + 2H_2O(l) + 4e^- \longrightarrow 4OH^-(aq)$$

The overall equation is:

$$2H_2(g) + O_2(g) \longrightarrow 2H_2O(l)$$

[1] Platinum can be coated on each electrode to catalyze the reaction.

Reference Website

Refer to the following websites for animations illustrating how a fuel cell works:

- http://www.lanl.gov/orgs/mpa/mpa11/animation.htm (accessed July 2014)
- http://www.hk-phy.org/energy/commercial/print/cell_is_print_e.html (accessed July 2014)

Advantages of hydrogen-oxygen fuel cells

Advantages of hydrogen-oxygen fuel cells include:

DSE 2012 Paper 1A Q18 DSE 2013 Paper 1B Q10(c)(ii)

- They do not emit air pollutants. Only water is produced.
- They have high efficiency of energy conversion. They convert chemical energy directly into electrical energy, achieving a remarkable efficiency of about 70%[2].

DSE 2012 Paper 1A Q18

- They can operate continuously if the flow of $H_2(g)$ and $O_2(g)$ can be maintained.

ASL 2008 Paper 1 Q3
AL 2008 Paper 1 Q2(b)

[2] In comparison, modern power plants and petrol engines using fossil fuels have a conversion efficiency from chemical energy to electrical energy or kinetic energy of about 40% only.

Limitations of hydrogen-oxygen fuel cells

There are several limitations of hydrogen-oxygen fuel cells.

- The materials used to make the electrodes are expensive.
- Manufacturing of these cells involves the production of toxic by-products.
- High-pressure tanks are needed to store the hydrogen for operating the cells (Fig. 21.8). There are safety concerns.
- The hydrogen needed can only be produced cheaply by using fossil fuels. N7

DSE 2013 Paper 1B Q10(c)(i)

- These cells do not work well at low temperatures. If the temperature falls much below 0 °C, the cell 'freezes'.

N7 Hydrogen-oxygen fuel cells may not be environmentally friendly if the hydrogen used is produced from fossil fuels / electricity is consumed in the production of hydrogen from water.



Fig. 21.8 Hydrogen refuelling stations provide hydrogen under pressure

Do you know

Direct methanol fuel cell

Direct methanol fuel cell (DMFC) uses methanol directly as the fuel. The liquid methanol (CH_3OH) is oxidized in the presence of water at the negative electrode, generating carbon dioxide, hydrogen ions and electrons. The electrons travel through the external circuit.

The hydrogen ions travel through the electrolyte. They react with oxygen from the air and the electrons from the external circuit to form water at the positive electrode.

The overall cell reaction is:

$2CH_3OH(l) + 3O_2(g) \longrightarrow 2CO_2(g) + 4H_2O(l)$

Manufacturers expect that DMFC will be widely used to power mobile electronic appliances in places where electric sockets are not available.

Worksheet Assessment Form

A worksheet and an assessment form on this activity are available for download from *Jing Kung Chemistry Website*.

Decision Making

Lithium or hydrogen powered vehicles

A parcel delivery company plans to replace their present fleet with either lithium or hydrogen powered vehicles. The manager invites two vehicle companies to present to the board of directors the features of their products.

- Company A sells vehicles powered by lithium ion cells.
- Company B sells vehicles powered by hydrogen-oxygen fuel cells.

Tasks

Your teacher will divide the class into groups.

Groups as vehicle company representatives

Two groups will act as representatives of the vehicle companies. Each group will give a 4-minute presentation on the following aspects:

1 the advantages of lithium ion cells or hydrogen-oxygen fuel cells;

2 the initial cost and the ongoing cost of repairs and maintenance;

3 the environmental impact of the vehicle.

Other groups

Other groups will be directors of the parcel delivery company. You need to

1 raise questions in the meeting;

2 decide whether to use lithium or hydrogen powered vehicles and justify your decision.

Refer to Teacher's Resources Book pages 50.

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

Chemistry Magazine

Oxygen absorbers for packaged foods

Oxygen causes rancid odours and changes in appearance and taste in packaged food. An oxygen absorber is needed to absorb oxygen and extend the shelf life of a food product.

An oxygen absorber is a small packet of material used to prolong the shelf life of food (Fig. 21.9). Oxygen absorbers usually contain iron powder which reacts with oxygen in the air, causing the iron powder to rust. When all the iron powder has been oxidized, the absorbing action stops.

The ingredients of an oxygen absorber are normally enclosed in a packet made of high-density polythene film or plastic-laminated paper.



Fig. 21.9 Packets of oxygen absorber

Questions

1 What type of packaged foods may have a packet of oxygen absorber?

2 Is the packet of oxygen absorber air-tight? Explain your answer.

3 On the label of one packet of oxygen absorber, it states 'DO NOT MICROWAVE'. From your daily life experiences, what kind of substances should not be heated in microwave ovens?

Refer to Teacher's Resources Book page 53.

Have you mastered?

Key terms

anode 陽極 84 cathode 陰極 84 inert electrode 惰性電極 87 fuel cell 燃料電池 90

hydrogen-oxygen fuel cell 氫氧燃料電池 90

Checklist

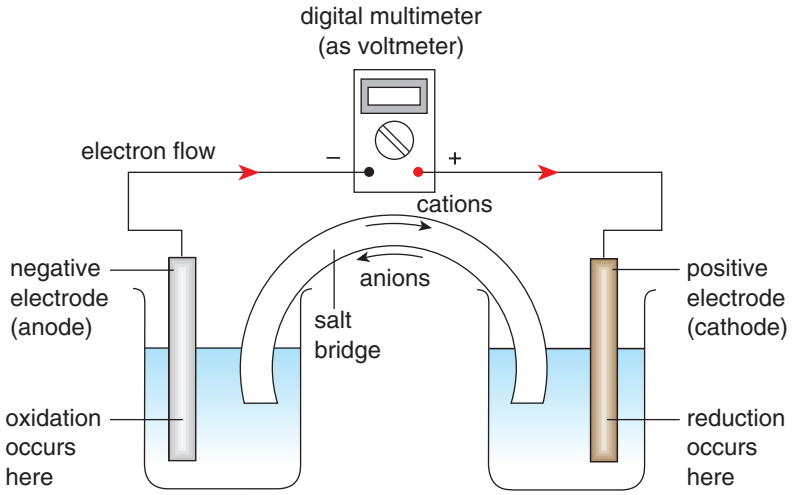
After studying this unit, you should be able to

- ☐ describe the structure of a zinc-carbon dry cell;
- ☐ write the overall equation for the reaction in a zinc-carbon dry cell;
- ☐ describe chemical cells with inert electrodes;
- ☐ predict the chemical changes that occur in chemical cells with inert electrodes;
- ☐ understand the working principle of a hydrogen-oxygen fuel cell;
- ☐ write the overall equation for the reaction in a hydrogen-oxygen fuel cell;
- ☐ state the pros and cons of a hydrogen-oxygen fuel cell.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 In a chemical cell, electrons flow in the external circuit and ions flow in the solutions.



	Negative electrode (anode)	Positive electrode (cathode)
Electrode where	oxidation occurs	reduction occurs
Direction of electron flow	from the negative electrode to the positive electrode in the external circuit	

2 In a zinc-carbon chemical cell:

a) At the negative electrode (or anode):

$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$

b) At the positive electrode (or cathode):

$2NH_4^+(aq) + 2e^- \longrightarrow 2NH_3(aq) + H_2(g)$

3 In a hydrogen-oxygen fuel cell, hydrogen and oxygen react to generate electricity.

a) At the negative electrode (or anode) compartment:

$H_2(g) + 2OH^-(aq) \longrightarrow 2H_2O(l) + 2e^-$

b) At the positive electrode (or cathode) compartment:

$O_2(g) + 2H_2O(l) + 4e^- \longrightarrow 4OH^-(aq)$

c) The overall equation is:

$2H_2(g) + O_2(g) \longrightarrow 2H_2O(l)$

Have you mastered?

Unit Exercise

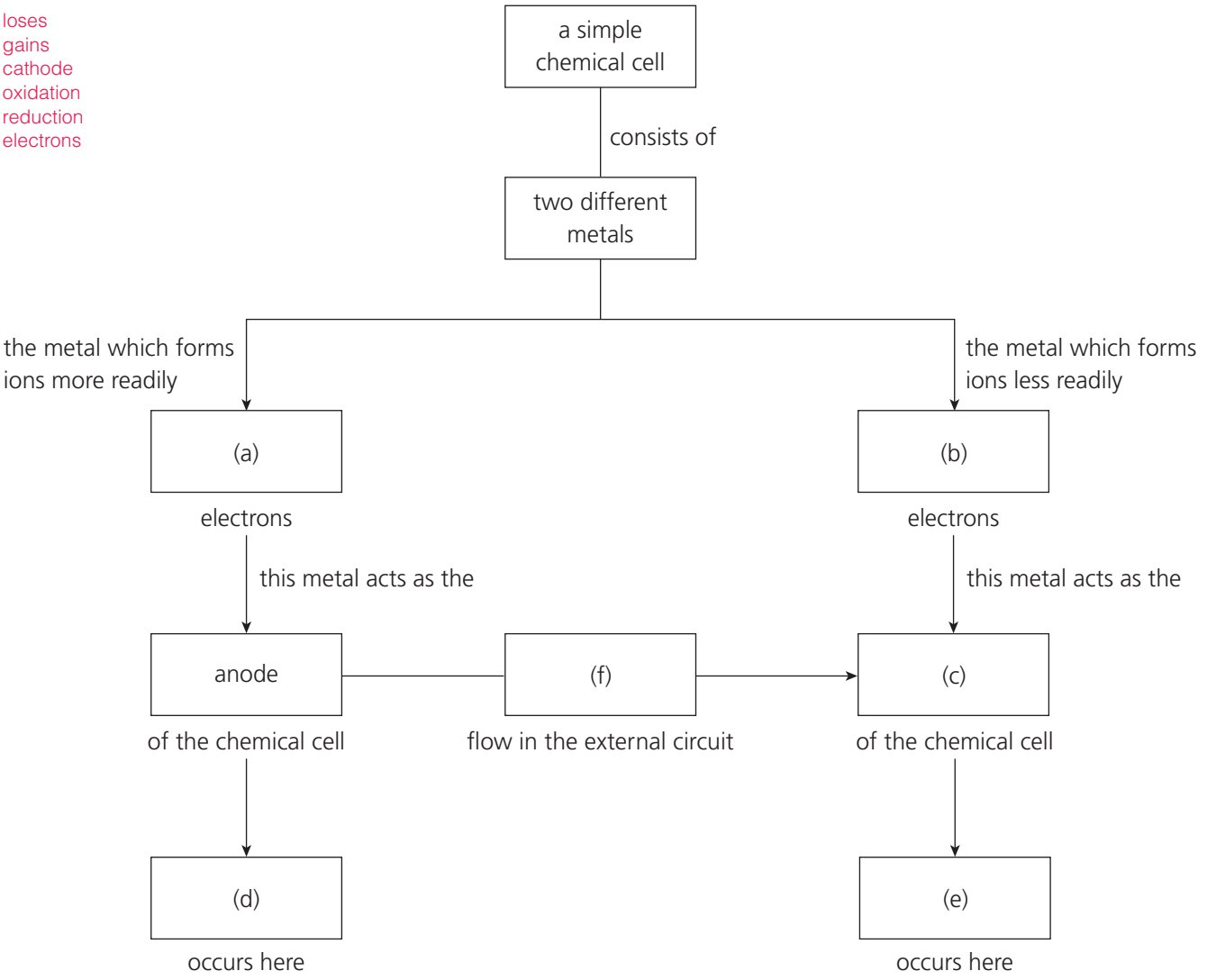
Refer to Teacher's Resources Book pages 53–55.

Note: The symbol ! indicates the level of difficulty of a question.

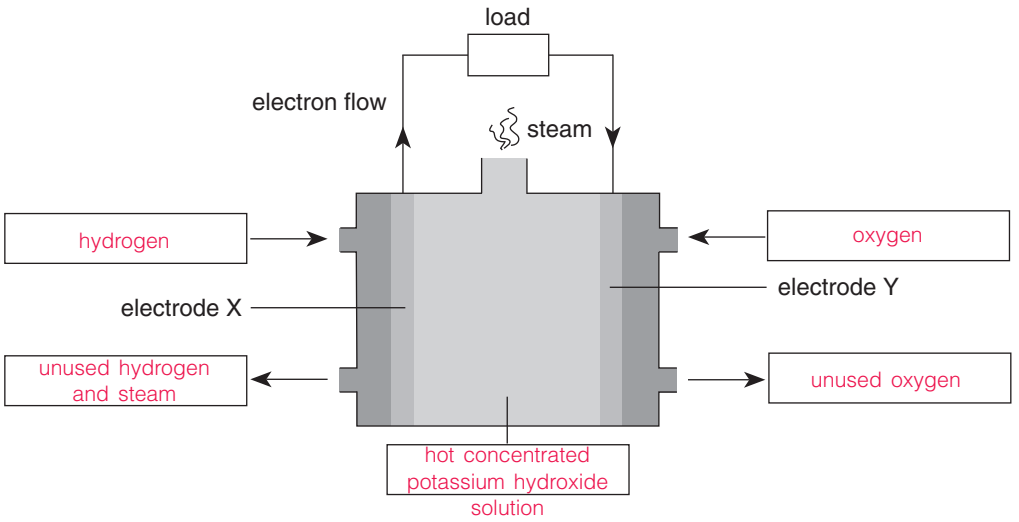
Part I Knowledge and understanding

1 Complete the following concept map.

a) loses
b) gains
c) cathode
d) oxidation
e) reduction
f) electrons



2 a) Complete the diagram for a hydrogen-oxygen fuel cell.



b) Write ionic half-equations for the reactions that occur at electrodes X and Y respectively.

Reaction at electrode X: $H_2(g) + 2OH^-(aq) \longrightarrow 2H_2O(l) + 2e^-$

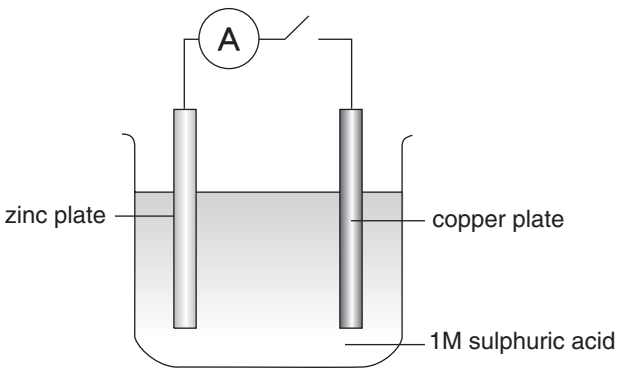
Reaction at electrode Y: $O_2(g) + 2H_2O(l) + 4e^- \longrightarrow 4OH^-(aq)$

c) Electrode X is the anode of the fuel cell.

Part II Multiple choice questions

Refer to Teacher's Resources Book for explanations.

Directions: Questions 3 and 4 refer to the following diagram.



3 Which of the following observations can be made in the above set-up?

A There is no observable change.
(B) Gas bubbles appear on the zinc plate.
C Gas bubbles appear on the copper plate.
D The sulphuric acid gradually turns blue.

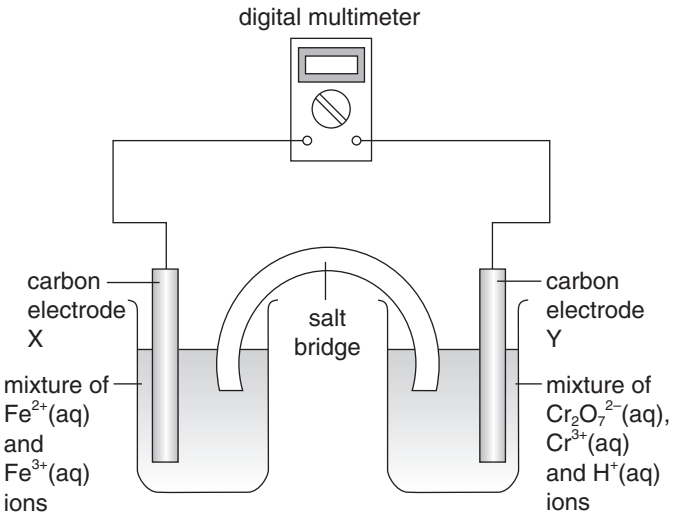
(HKCEE, Paper 2, 2009, 8)

4 What will occur when the circuit is closed?

A Both metal plates gradually dissolve.
B The sulphuric acid gradually turns blue.
(C) The hydrogen ions in the solution are reduced to hydrogen gas.
D Electrons flow from the copper plate to the zinc plate in the external circuit.

(HKCEE, Paper 2, 2009, 9)

5 Consider the chemical cell shown below:



Which of the changes represented by the following ionic half-equations would occur if the cell is producing a current?

A $Fe^{2+}(aq) + 2e^- \longrightarrow Fe(s)$
B $Fe^{3+}(aq) + e^- \longrightarrow Fe^{2+}(aq)$
C $2Cr^{3+}(aq) + 7H_2O(l) \longrightarrow Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^-$
(D) $Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \longrightarrow 2Cr^{3+}(aq) + 7H_2O(l)$

6 The ionic half-equations below show changes that occur at the two electrodes, X and Y, of an alkaline manganese cell during discharge.

At electrode X: $Zn(s) + 2OH^-(aq) \longrightarrow ZnO(s) + H_2O(l) + 2e^-$

At electrode Y: $2MnO_2(s) + H_2O(l) + 2e^- \longrightarrow Mn_2O_3(s) + 2OH^-(aq)$

Which of the following statements about the cell is correct?

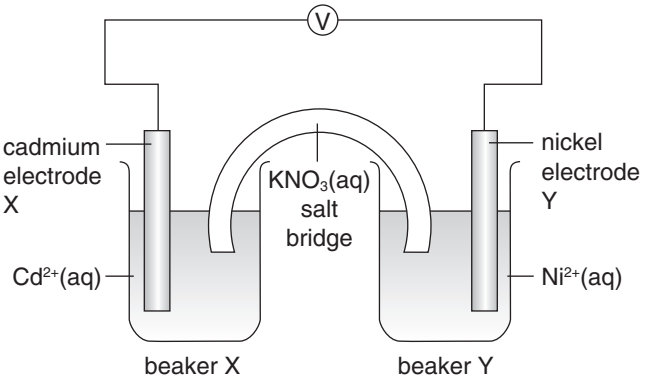
A Electrode X acts as the cathode.
(B) Electrode Y acts as the positive electrode.
C Reduction occurs at electrode X.
D Electrons flow from electrode Y to electrode X in the external circuit.

7 Which of the following would NOT occur when a zinc-carbon cell is supplying electricity?

A Water is produced.
B The zinc case becomes thinner.
C Ammonium ions are consumed.
(D) Manganese compound is oxidized.

(HKCEE, Paper 2, 2009, 34)

8 Consider the chemical cell shown below. As time goes by, the green colour of the solution in beaker Y gradually fades out.



Which of the following statements about the cell are correct?

(1) Electrons flow from electrode X to electrode Y in the external circuit.
(2) Reduction occurs at electrode Y.
(3) $NO_3^-(aq)$ ions migrate from the salt bridge into the nickel half-cell.

(A) (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

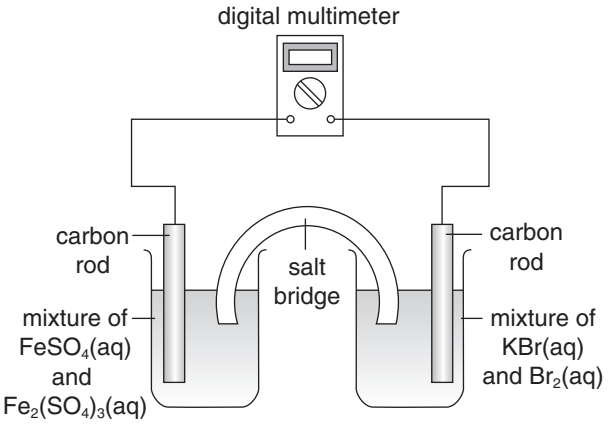
9 Which of the following statements concerning a hydrogen-oxygen fuel cell is / are correct?

(1) It produces non-polluting product.
(2) The membrane in it selectively allows hydroxide ions to pass through.
(3) It can continuously produce electricity as long as hydrogen and oxygen are supplied under operating conditions.

A (1) only
B (2) only
(C) (1) and (3) only
D (2) and (3) only

(HKDSE, Paper 1A, 2012, 18)

10 The following diagram shows the set-up of a chemical cell.



Given that $Br_2(aq)$ is a stronger oxidizing agent than $Fe^{3+}(aq)$, which of the changes represented by the following ionic half-equations would occur if the cell is producing a current?

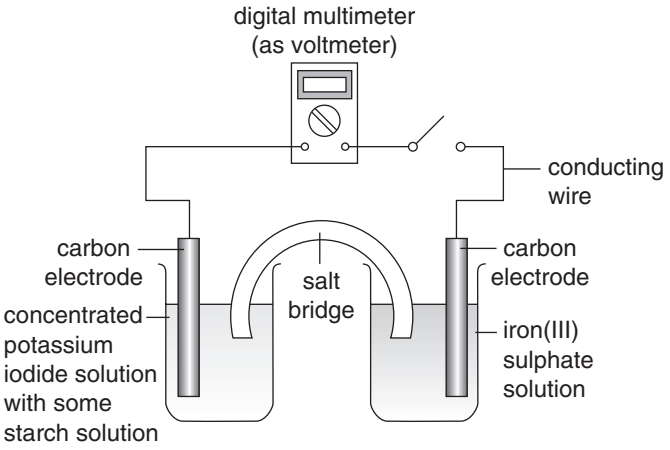
(1) $Fe^{3+}(aq) + e^- \longrightarrow Fe^{2+}(aq)$
(2) $Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq) + e^-$
(3) $2Br^-(aq) \longrightarrow Br_2(aq) + 2e^-$

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKCEE, Paper 2, 2008, 44)

Part III Structured questions

11 When the circuit in the set-up shown below was closed, the colour of iron(III) sulphate solution changed from yellow-brown to green gradually.



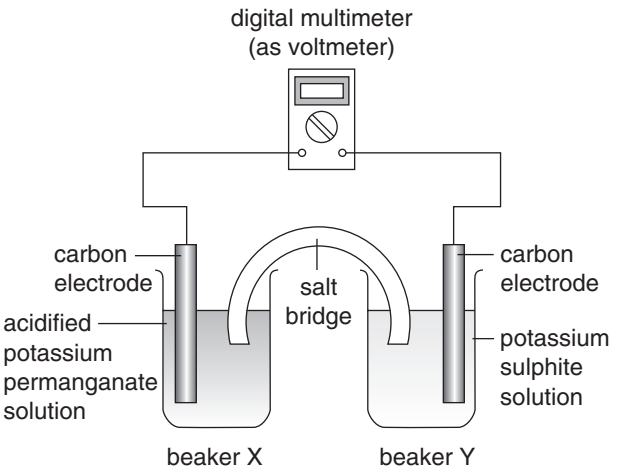
a) Explain the colour change of the iron(III) sulphate solution. Write an ionic half-equation for the reaction that occurred.

b) i) Explain what would be observed in the potassium iodide solution after some time. Write an ionic half-equation for the reaction that occurred.

ii) Explain whether the iodide ions were oxidized or reduced.

c) Identify the direction of electron flow in the external circuit.

12 Consider the chemical cell shown below:



It is known that permanganate ion has a higher tendency of being reduced than sulphite ion.

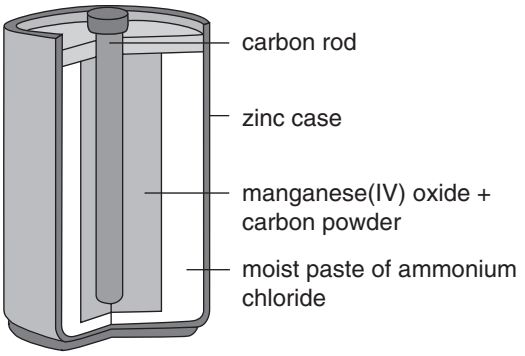
a) Suggest the observable change for the acidified potassium permanganate solution after some time. Write an ionic half-equation for the reaction that occurs.

b) i) Write an ionic half-equation for the change that occurs for the potassium sulphite solution.

ii) Explain whether the sulphite ions are oxidized or reduced.

c) Identify the direction of electron flow in the external circuit.

13 The following diagram shows a zinc-carbon cell.



a) i) Which substance acts as the cathode?

ii) Write an ionic half-equation for the reaction that occurs at the cathode.

b) i) Which substance acts as the anode?

ii) Write an ionic half-equation for the reaction that occurs at the anode.

c) Explain the function of manganese(IV) oxide.

d) The voltage of the cell drops if a current is drawn from the cell rapidly. Explain briefly.

e) Explain why the zinc case of a used cell is thinner than that of a new cell.

f) Explain why disposal of 'flat' cells presents a pollution problem.

g) Explain why zinc-carbon cells should be removed from electric devices when not in use for a long period.

14 Silver oxide button cells are commonly used instead of mercury button cells since the chemicals used are not as toxic. The following ionic half-equations represent the chemical changes that occur in a silver oxide button cell.

At electrode X: $Ag_2O(s) + H_2O(l) + 2e^- \longrightarrow 2Ag(s) + 2OH^-(aq)$

At electrode Y: $Zn(s) + 2OH^-(aq) \longrightarrow Zn(OH)_2(s) + 2e^-$

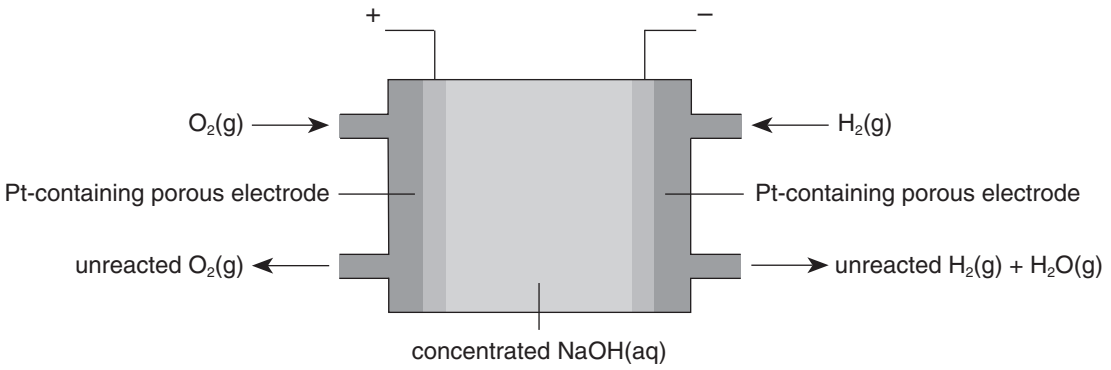
a) In which direction do electrons flow in the external circuit, from X to Y or Y to X?

b) Decide and explain which electrode, X or Y, is the cathode of the cell.

c) Write an ionic equation for the discharging of the silver oxide button cell.

15 The diagram below shows the design of a hydrogen-oxygen fuel cell. Its working principle is based on the overall reaction:

$2H_2(g) + O_2(g) \longrightarrow 2H_2O(g)$



a) For the fuel cell, write ionic half-equations for the cathodic and anodic reactions.

b) Suggest TWO advantages of using fuel cells over using other conventional electrochemical cells.

c) Suggest why fuel cells are NOT widely used in daily life despite their having a number of advantages.

(HKASLE, Paper 1, 2008, 3)